

Optimising a Scarce Study Budget: A Linear Programming Approach to Undergraduate Academic Workload

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Abstract. A survey of 147 third-year Mathematics students at the University of Chittagong found that the typical student has roughly 7.6 hours of self-study time per week, fewer than 46 minutes per course. This project models that allocation as a Linear Programme and solves it via the Simplex method. Two formulations are compared: a Baseline weighted by course credits and an Improved version that scales those weights by a five-level difficulty coefficient. The Improved model produces five distinct, individually justifiable weekly allocations and sensitivity analysis yields a shadow price of 3.0, identifying precisely where an extra hour of study is most productively spent.

1. Introduction

Linear programming is often taught as a theoretical exercise, but its true utility lies in managing real-world scarcity. For a mathematics student at the University of Chittagong, this scarcity is felt acutely when attempting to balance twelve papers within a 1,000-mark syllabus[4]. To move beyond habit-based time allocation, this project formalises the weekly study schedule as a **Linear Programme (LP)**.

At its core, this is a resource allocation problem: a fixed budget, competing demands and the need to make every unit count. Students face this every week, yet almost no one approaches it with the rigour the problem actually deserves.

A central concept utilized here is the **shadow price**, the marginal gain obtained by relaxing a binding constraint by one unit. For a student, this provides a precise, mathematical answer to where one additional hour of effort is most productive once a weekly budget is exhausted. To ensure the model reflects reality, a primary survey of 147 students from the 2021–22 and 2022–23 sessions was conducted. By solving this model via the Simplex method and **implementing the solution computationally**, I move from intuitive planning to a data-driven framework that optimizes academic

return under strict time constraints.

2. Survey, Follow-Up and Budget Derivation

2.1. Survey Design

The survey covered third-year B.Sc. (Hons.) Mathematics students from sessions 2021–22 and 2022–23 at the University of Chittagong. Since the department revises its curriculum every two years, these two cohorts are the entire population to whom the current syllabus applies ($N = 200$). Of these, $n = 147$ responded, giving a response rate of 73.5 %.

The question was straightforward: “*On average, how many hours do you study per day outside of class?*” Respondents chose one of five options: *Less than 30 minutes*, *30 minutes to 1 hour*, *2 to 3 hours*, *4 hours or above*, or *Only before exams*.

2.2. Follow-Up: Correcting the Open-Ended Category

The *4 hours or above* option has no natural midpoint. To avoid relying on an arbitrary estimate, all 11 students who chose this option were contacted individually and asked for their actual daily study time. Their responses were:

- 2 students: 6 hours per day
- 4 students: 5 hours per day
- 5 students: 4 hours per day

The corrected midpoint, calculated directly from these responses, is:

$$m_{4h+} = \frac{2 \times 6 + 4 \times 5 + 5 \times 4}{11} = \frac{52}{11} \approx 4.727 \text{ h/day} \quad (1)$$

This ensures every number in the budget calculation comes from something a real student actually reported.

2.3. Results and Budget Derivation

Response Category	Count	%	Midpoint (h/day)
Less than 30 minutes	7	4.8%	0.2500
30 minutes to 1 hour	38	25.9%	0.7500
2 to 3 hours	31	21.1%	2.5000
4 hours or above*	11	7.5%	4.7273
Only before exams	60	40.8%	0.0000
Total	147	100%	—

Table 1: Daily self-study distribution ($n = 147$, response rate 73.5%, sessions 2021–22 and 2022–23). The *Only before exams* category is the largest single group at 40.8%, showing that a substantial share of students have no consistent study habit during term. Those students are assigned 0 h/day. The midpoint for the open-ended *4 hours or above* category was determined to be 4.727 h/day, a figure established directly through individual verification of student responses.

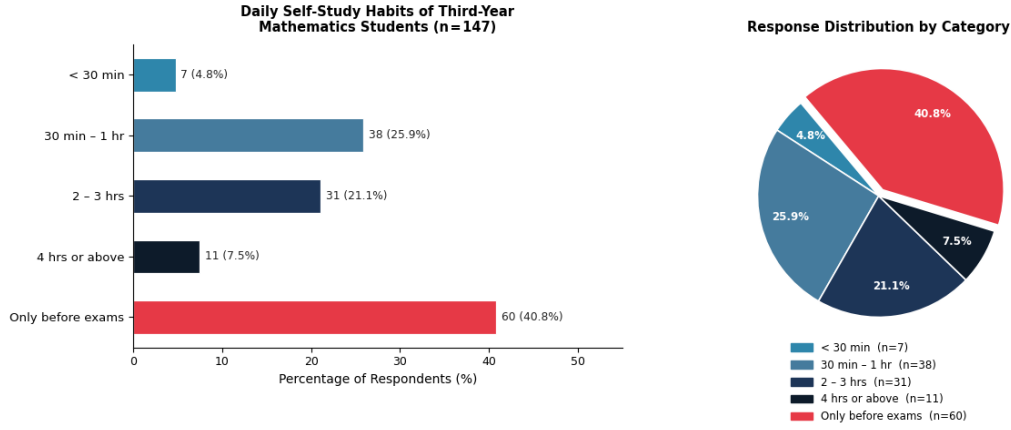


Figure 1: Survey response distribution as a horizontal bar chart (left) and pie chart (right). The *Only before exams* group dominates at 40.8%. The determined midpoint of 4.727 h/day for the open-ended category is shown in the chart label.

Hence, the weighted mean daily study time is:

$$\bar{h}_{\text{daily}} = \frac{\sum_{k=1}^5 n_k m_k}{n} = \frac{7(0.25) + 38(0.75) + 31(2.5) + 11(4.727) + 60(0)}{147} = \frac{159.75}{147} \approx 1.087 \text{ h/day} \quad (2)$$

Multiplying by seven gives the weekly budget used throughout this project:

$$\text{LP Budget} = 7.6 \text{ h/week} \quad (i.e. 7 \text{ h } 36 \text{ min})$$

Spread across ten courses, that is under 46 minutes per course per week, which is genuine scarcity that makes the optimisation both non-trivial and practically useful.

3. Model Formulation

3.1. Decision Variables

Let $x_i \in \mathbb{R}_{\geq 0}$ be the weekly self-study hours allocated to course i , for $i = 1, \dots, 10$, as defined in the official curriculum [4]. Math 311 (Sessional (Tutorial+Attendance)) and Math 312 (Viva Voce) are excluded since their marks mainly depend on attendance and oral performance, neither of which responds meaningfully to desk study hours.

x_1 : Math 301 — Real Analysis (II)

x_6 : Math 306 — Mechanics

x_2 : Math 302 — Complex Analysis (I)

x_7 : Math 307 — Numerical Analysis (I)

x_3 : Math 303 — Abstract Algebra

x_8 : Math 308 — Linear Programming

x_4 : Math 304 — Partial Differential Equations

x_9 : Math 309 — Number Theory

x_5 : Math 305 — Methods of Applied Mathematics

x_{10} : Math 310 — Mathematical Computing Lab

3.2. Weighting Strategy

The model utilizes course credits (c_i) as the fundamental baseline for time allocation. Since every course in this programme maintains a constant ratio of 25 marks per credit ($M_i/c_i = 25$), credits alone provide an objective measure of the university's academic valuation. Course names, credit allocations and mark distributions are sourced from the official *Curriculum: B. Sc.(Hon's) in Mathematics (Sessions 2021–2022 & 2022–2023)* provided by the Department of Mathematics, University of Chittagong [4]. This baseline ensures that 4-credit courses are prioritized over 3-credit ones, creating a neutral starting point for the optimization.

To capture the varying cognitive demands of the subjects, I have introduced a difficulty multiplier d_i . This multiplier refines the baseline weight using a symmetric five-level scale:

- **Most Difficult:** $d_i = 1.2$
- **Difficult:** $d_i = 1.1$
- **Medium:** $d_i = 1.0$
- **Easy:** $d_i = 0.9$
- **Easiest:** $d_i = 0.8$

The 0.1 spacing between levels ensures that each step represents an identical proportional change in priority. By keeping the multipliers close to 1.0, the model preserves credits as the primary driver while allowing difficulty to act as a secondary adjustment. The specific assignments (shown in Table 2) are based on personal academic experience

and peer discussions within the department. This approach is intentionally flexible. Any student using this framework can reassign courses without changing the model structure.

Course	M_i	c_i	d_i	$c_i \cdot d_i$	Reason
Math 301 — Real Analysis (II)	75	3	1.0	3.0	Rigorous, but builds on 2nd year foundations
Math 302 — Complex Analysis (I)	75	3	1.0	3.0	Theoretical; well-structured and progressive
Math 303 — Abstract Algebra	100	4	1.2	4.8	Entirely proof-based; highest abstraction
Math 304 — PDEs	75	3	1.0	3.0	Conceptually rich; manageable with practice
Math 305 — Methods of Applied Mathematics	100	4	1.1	4.4	Broad advanced techniques; genuine depth required
Math 306 — Mechanics	100	4	1.2	4.8	Rigorous blend of mathematics and physics
Math 307 — Numerical Analysis (I)	75	3	0.8	2.4	Algorithmic; procedure is clearly defined
Math 308 — Linear Programming	100	4	0.8	3.2	Structured Simplex method; framework is familiar
Math 309 — Number Theory	75	3	0.9	2.7	Abstract but self-contained and approachable
Math 310 — Mathematical Computing Lab	75	3	0.8	2.4	Practical and guided; lowest abstraction
Math 311 — Sessional	75	3	—	—	Excluded (attendance-based)
Math 312 — Viva Voce	75	3	—	—	Excluded (oral examination)
Total	1000	40	—	—	

Table 2: Third-year syllabus (2021–22 and 2022–23 sessions) with difficulty multipliers and composite weights. Note: The biennial update cycle ensures these difficulty assignments are strictly relevant to the current curriculum structure.

3.3. Objective Functions

Following the standard LP maximisation framework [2], two objective functions are defined.

Baseline — weighted by credits only:

$$\max Z_{\text{base}} = \sum_{i=1}^{10} c_i \cdot x_i \quad (3)$$

Improved — weighted by credits times difficulty:

$$\max Z_{\text{imp}} = \sum_{i=1}^{10} c_i \cdot d_i \cdot x_i \quad (4)$$

3.4. Constraints

C1. Weekly budget — total weekly study time cannot exceed the figure derived from the survey (Equation 2):

$$\sum_{i=1}^{10} x_i \leq 7.6 \quad (5)$$

C2. Minimum coverage — every subject must receive at least 20 minutes ($\frac{1}{3}$ h) per week, so none is entirely neglected. The mandatory portion is $10 \times \frac{1}{3} \approx 3.33$ h:

$$x_i \geq \frac{1}{3}, \quad \forall i \quad (6)$$

C3. Proportional upper bound — each subject is capped at 1.3 times its fair-share allocation, preventing the solver from directing all surplus into the single highest-weighted course [2]:

$$x_i \leq \frac{w_i}{\sum_j w_j} \times 7.6 \times 1.3, \quad \forall i \quad (7)$$

where $w_i = c_i$ (Baseline) or $w_i = c_i d_i$ (Improved).

C4. Tie-breaking equality — when two courses share the same objective weight ($w_i = w_j$), the constraint $x_i = x_j$ is imposed. Without this the LP is degenerate: any reallocation between equal-weight courses leaves the objective unchanged, making the solver's choice arbitrary [2]. Forcing equality gives the unique symmetric solution:

$$x_i = x_j, \quad \forall i, j \text{ such that } w_i = w_j \quad (8)$$

C5. Non-negativity — Standard for all decision variables:

$$x_i \geq 0, \quad \forall i \quad (9)$$

The resulting model consists of 10 decision variables and 21 primary linear constraints, providing a structured foundation for computational implementation.

4. Results and Discussion

The optimization models were implemented in Python using the **PuLP** library [3] and solved via the CBC solver. The Simplex algorithm [1] reached optimality for both formulations in under 0.005 seconds. Table 3 reports the weekly allocations; Figure 2 shows the comparison.

Course	Baseline	Improved
Math 301 — Real Analysis (II)	30 min	31 min
Math 302 — Complex Analysis (I)	30 min	31 min
Math 303 — Abstract Algebra	1 h 10 m	1 h 24 m
Math 304 — PDEs	30 min	31 min
Math 305 — Methods of Applied Mathematics	1 h 10 m	1 h 17 m
Math 306 — Mechanics	1 h 10 m	1 h 24 m
Math 307 — Numerical Analysis (I)	30 min	20 min
Math 308 — Linear Programming	1 h 10 m	56 min
Math 309 — Number Theory	30 min	20 min
Math 310 — Mathematical Computing Lab	30 min	20 min
Math 311 — Sessional	<i>Attendance-based</i>	
Math 312 — Viva Voce	<i>Oral examination</i>	
Total	7 h 36 m	7 h 36 m

Table 3: Optimal weekly allocations from both models (credit and difficulty data for each course are in Table 2). Both solutions exhaust the full 7.6-hour budget. The Baseline assigns equal time within each credit group: 1 h 10 m for all 4-credit courses and 30 min for all 3-credit courses, with no distinction made within those groups, so Abstract Algebra and Linear Programming receive identical time despite being at opposite ends of the difficulty spectrum. The Improved model produces five distinct levels where every allocation traces directly to the composite weight $c_i \cdot d_i$ in Table 2.

4.1. Sensitivity Analysis and the Shadow Price

The 7.6-hour budget is **strictly binding** in both solutions; every available minute is allocated. Sensitivity analysis [2] lets us ask the most useful question of the project: *if I could find one extra hour per week, where should it go and how much would it help?*

The answer is the shadow price λ . In the Improved solution, Abstract Algebra and Mechanics have both reached their proportional upper bounds. The next courses with remaining capacity are Real Analysis, Complex Analysis and PDEs, each carrying composite weight $c_i \cdot d_i = 3 \times 1.0 = 3.0$. The shadow price of the budget constraint is

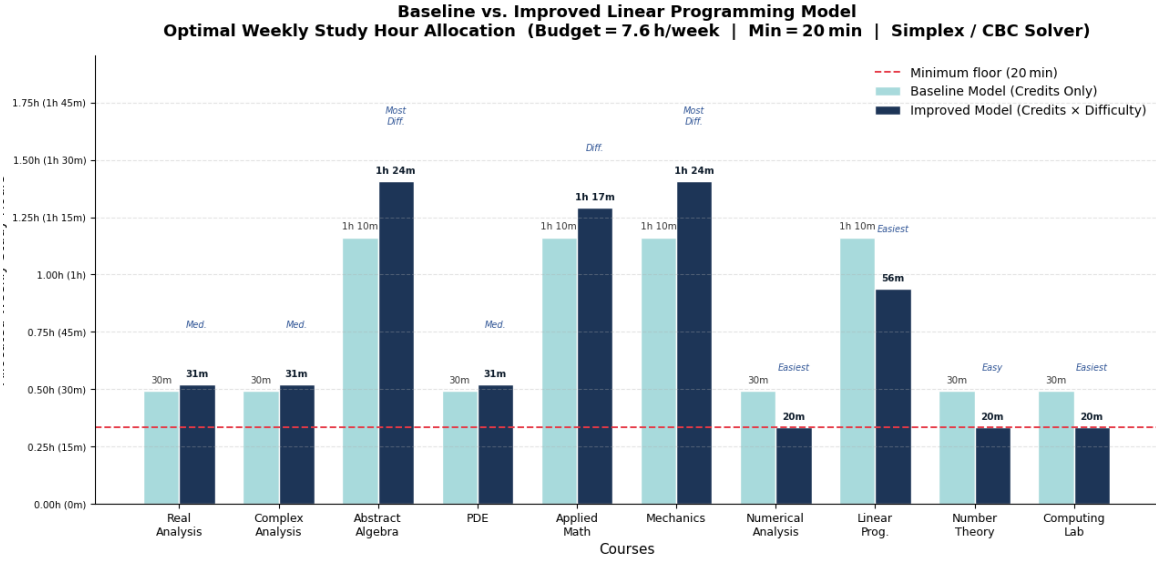


Figure 2: Baseline (light bars) vs. Improved (dark bars) weekly allocations. Difficulty levels are annotated above each Improved bar; the dashed red line marks the 20-minute minimum floor. Linear Programming drops from 1 h 10 m to 56 min because its composite weight (3.2) falls below the Medium 3-credit level (3.0). Numerical Analysis, Number Theory, and the Computing Lab all reach the floor once the budget is absorbed by higher-priority courses.

therefore:

$$\lambda = 3.0 \quad (10)$$

Each extra hour directed optimally returns 3.0 objective units. This is not the highest weight in the model, as Abstract Algebra sits at 4.8, but it is the highest weight among courses that can still absorb more time. The shadow price reflects where the constraint actually bites right now, not where unlimited time would ideally be spent.

5. Conclusion

This project started from a simple observation: with fewer than 46 minutes per subject per week, how a third-year Mathematics student at the University of Chittagong spreads study time is not a trivial decision. Linear programming [2] gave me a principled way to make that decision rather than drifting into familiar patterns.

The 7.6 h/week budget rests entirely on real data: a survey with a 73.5 % response rate covering the full target population, with a follow-up that replaced the open-ended *4 hours or above* category with an individually confirmed midpoint of 52/11 $\approx$ 4.73 h/day. Every parameter traces back to something observed.

The two formulations together make the value of difficulty weighting concrete. Credits alone produce a Baseline that is internally consistent but blind to academic demand within credit groups. The five-level difficulty multiplier resolves this, giving five distinct allocation levels where every decision has a clear and specific justification.

The shadow price of 3.0 (Equation 10) is the most directly useful result. A marginal hour of study directed to a Medium-level course such as Real Analysis, Complex Analysis or PDEs returns 3.0 objective units. That is a precise answer to the question this project set out to address.

Two extensions follow naturally from here. The first is **stochastic programming**: replacing the fixed 7.6 h budget with a distribution over weekly study time drawn from the survey's own response frequencies, which already exist and require no additional data collection. The second is a **multi-period formulation** where both the budget and difficulty weights shift week by week as semester pressure builds toward examinations. Both are grounded in the same empirical foundation built here and both represent the direction I intend to take this work.

References

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- [3] Mitchell, S., O'Sullivan, M., & Dunning, I. (2011). *PuLP: A Linear Programming Toolkit for Python*. University of Auckland. Available at: <https://optimization-online.org/2011/09/3178/>
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